Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Retail Sales & Performance Dashboard

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Retail Sales & Performance Dashboard

GITHUB Link: https://github.com/pragathidevanga/Business-Analytics/

Problem Statement

You are a Data Analyst in a retail company. Sales, product, and store data are maintained in a MySQL database. Management wants a centralized reporting system to track sales performance across products and store locations. The task is to design the database in MySQL, import the data into Power BI, establish the required relationships, and prepare an interactive dashboard for business reporting.

Methodology / Steps Followed

Step 1: Data Collection

- Product data collected: ProductID, Product Name, Category, Price, Supplier.
- Store data collected: StoreID, Store Name, City, State, StoreType.
- Sales transaction data collected: SaleID, ProductID, StoreID, SaleDate, Quantity, TotalAmount.
- Customer data collected separately: CustomerID, CustomerName, Gender, Age, PaymentMethod, CustomerSegment (linked via SaleID).
- Four distinct datasets gathered to support product, store, and customer-level analysis.

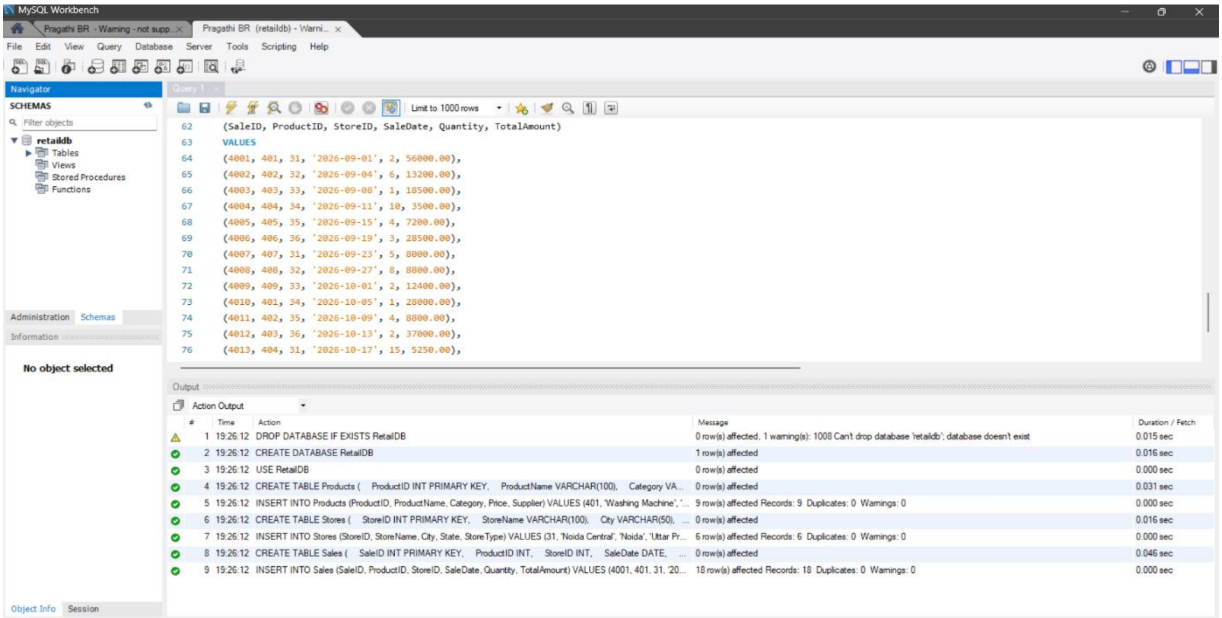

Fig 1: MySQL Workbench output showing successful creation of RetailDB database and insertion of Products, Stores, and Sales data.

Step 2: Data Import

- Created RetailDB database in MySQL Workbench (DROP DATABASE IF EXISTS → CREATE DATABASE RetailDB).
- Created Products table (ProductID as primary key) and inserted 9 records.
- Created Stores table (StoreID as primary key) and inserted 6 records.
- Created Sales table (SaleID as primary key, linked to ProductID & StoreID) and inserted 18 records.
- Verified successful execution in MySQL Workbench output log — 0 duplicates, 0 warnings.

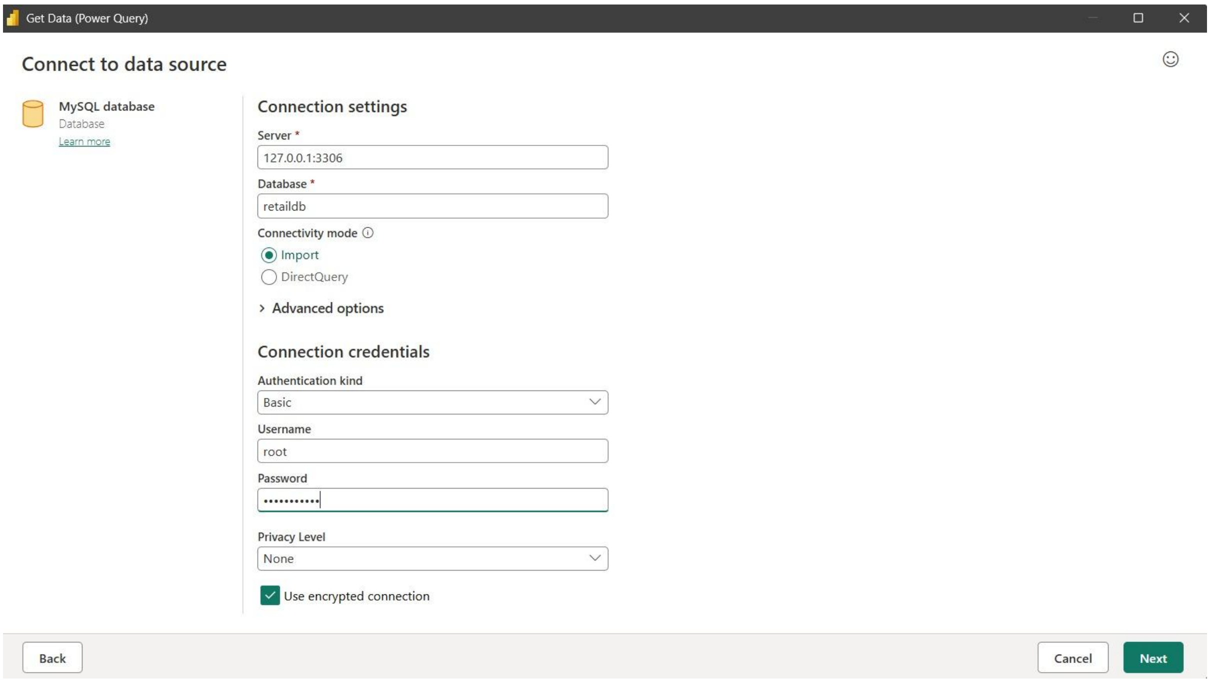

Fig 2: Power BI Get Data window showing connection to MySQL RetailDB and table selection."

- Opened Power BI Desktop and connected via Get Data → MySQL Database.
- Imported retaildb products, retaildb sales, and retaildb stores tables.
- Imported Customer Details separately via Get Data → Excel/CSV.

Step 3: Data Cleaning and Transformation

- Checked all four tables for duplicate records — none found.
- Checked key fields (ProductID, StoreID, SaleID, CustomerID) for missing values — none found.
- Verified/corrected data types: SaleDate as Date, TotalAmount/Price as Decimal, Age/Quantity as Whole Number.
- Renamed columns for clarity (e.g., "Product Name", "Store Name") to make reporting business-friendly.

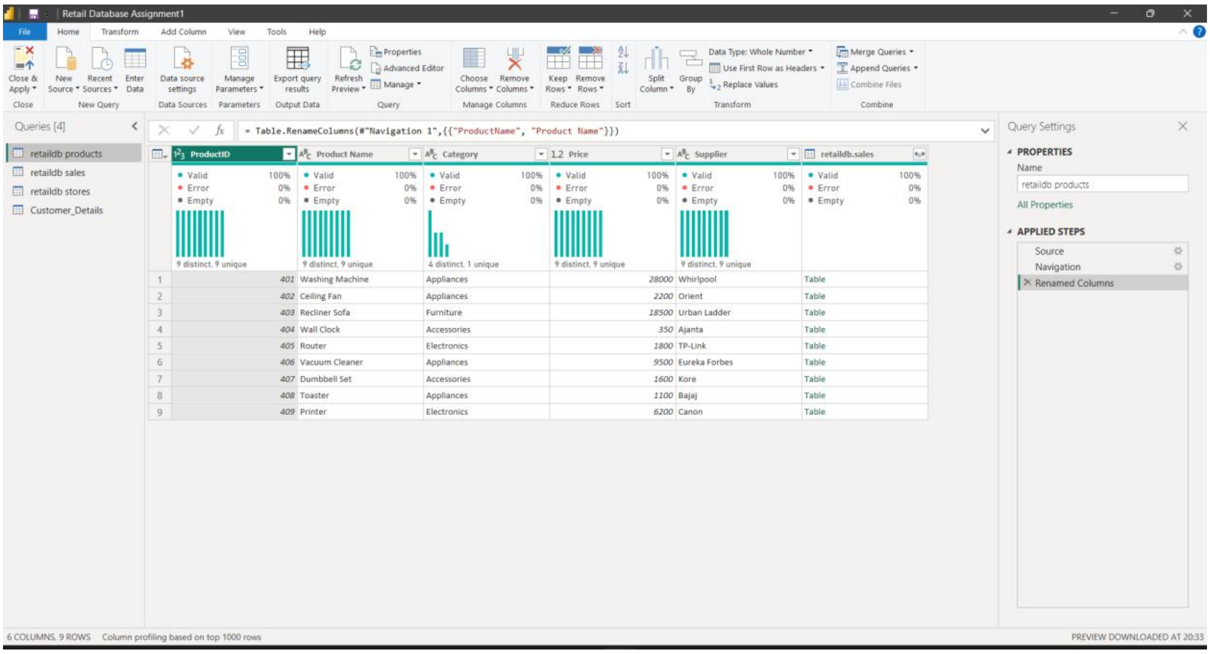

Fig 3 — Power Query Editor showing applied transformation steps (data type correction, duplicate check, and column renaming) on the retaildb sales table.

Step 4: Data Modeling

- Constructed the data model in Power BI's **Model view** after cleaning and validating all four tables.
- Created **Relationship 1**: retaildb sales ↔ retaildb products, linked via **ProductID** — enables tracing each sale back to its product, category, price, and supplier.
- Created **Relationship 2**: retaildb sales ↔ retaildb stores, linked via **StoreID** — enables sales analysis by city, state, and store type.
- Created **Relationship 3**: retaildb sales ↔ Customer Details, linked via **SaleID** — enables demographic slicing by gender, age, payment method, and customer segment.
- Verified all three relationships as **one-to-many**, with retaildb sales positioned as the "many" side (fact table) and Products, Stores, and Customer Details as the "one" side (dimension tables).
- Confirmed the structure resembles a **star schema**, ensuring accurate aggregation (SUM, COUNT) across dimensions without circular references or duplicate row multiplication.
- Validated the model to prevent downstream errors in KPIs and visuals before proceeding to dashboard development.

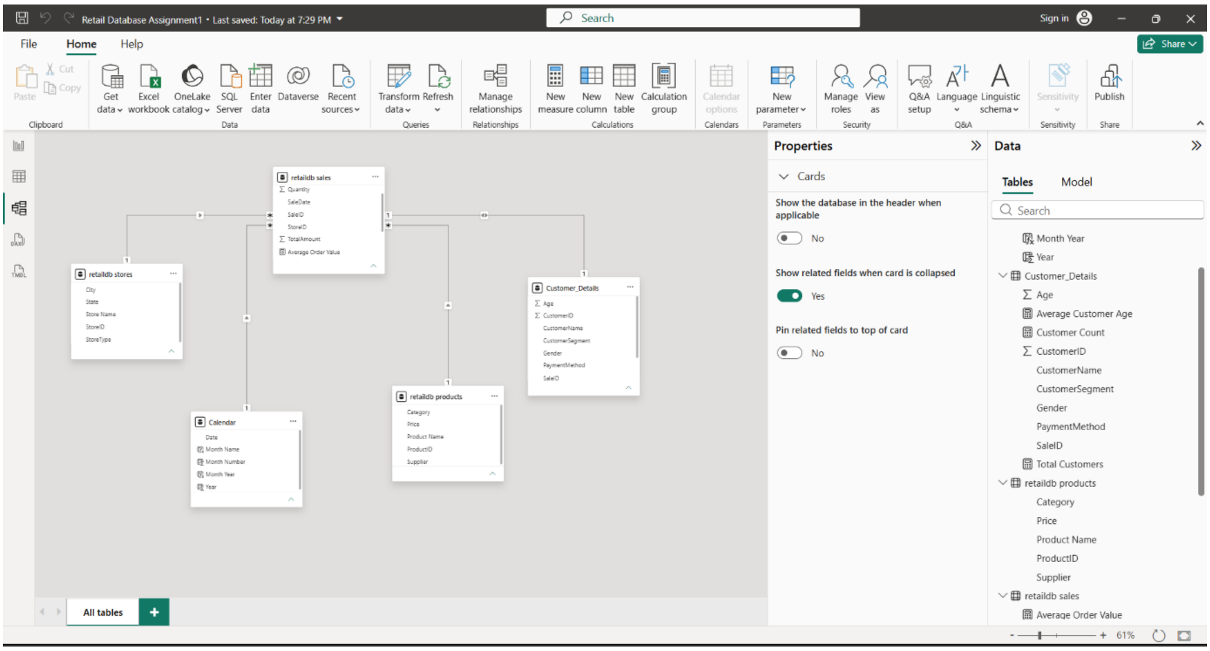

Fig — Power BI Model view showing one-to-many relationships between Sales, Products, Stores, and Customer Details.

Step 5: Dashboard Development

- Created core DAX measures:
    - **Total Sales** = SUM('retaildb sales'[TotalAmount])
    - **Total Orders** = COUNT('retaildb sales'[SaleID])
    - **Total Quantity** = SUM('retaildb sales'[Quantity])
    - **Total Customers = DISTINCTCOUNT('Customer_Details'[CustomerID])**

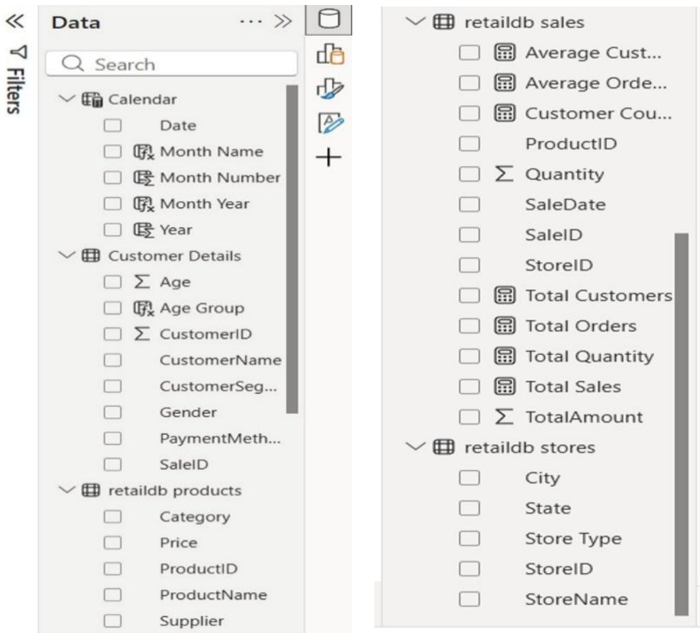

Fig — DAX formula Created in PowerBI

Step 5: Dashboard Development :

Created DAX measures: **Total Sales, Total Orders, Total Quantity, Total Customers, Average Order (value), Measure**.

- Built 4 KPI cards on Page 1: **Total Sales (300.25K), Total Orders (18), Total Quantity (86), Total Customers (18)**.
- Built a horizontal bar chart: **Total Sales by City** (Mangalore highest at 84K, followed by Noida, Ranchi, Thane, Guwahati, Vadodara).
- Built a column chart: **Retail Sales by Category** (Appliances leading at 168K, followed by Furniture, Electronics, Accessories).
- Built a line chart: **Retail Sales by Date** (Sep–Nov 2026 trend, peaking at 37K in late Oct).
- Added slicers for **Month Year** and **Category**, and a **City** filter list (Guwahati, Mangalore, Noida, Ranchi, Thane, Vadodara).
- Designed a navigation panel on Page 1 titled "Retail Dashboard" with sections: Overview, Map View, Product Insights, Customer Insights, Insights.

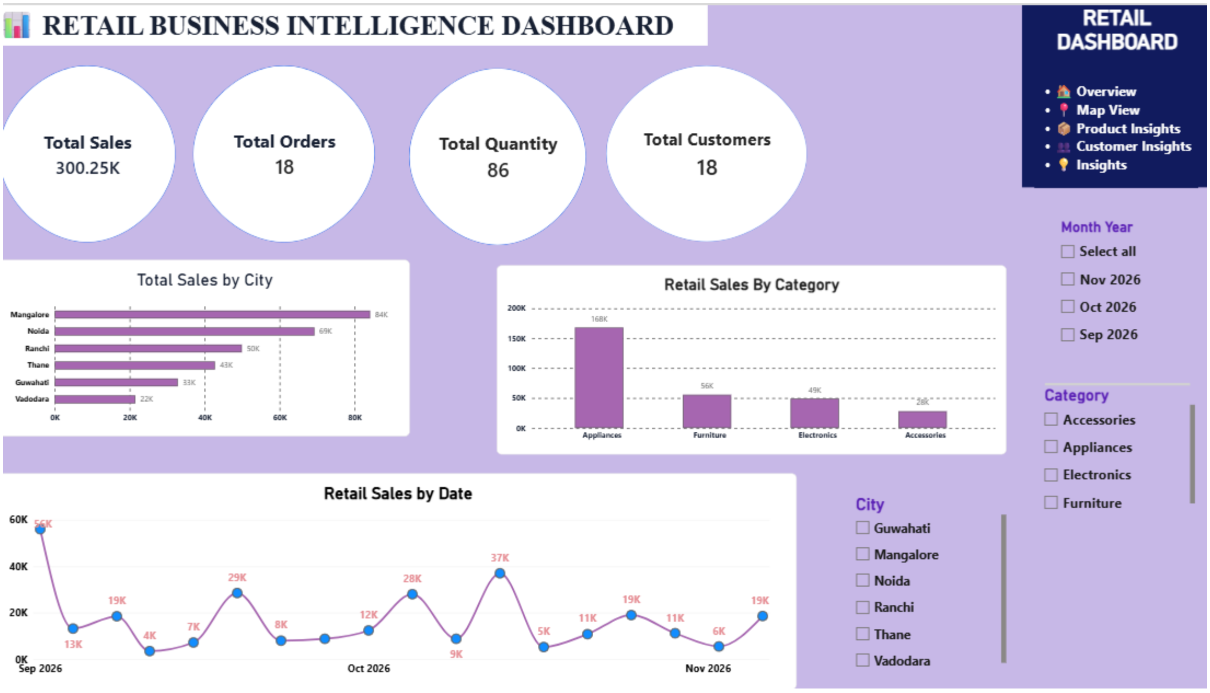

Fig — Retail Business Intelligence Dashboard (Page 1) showing KPI cards, Total Sales by City, Retail Sales by Category, and Retail Sales by Date, with Month Year and Category slicers.

- Built a second page (**Page 2**) with a **map visual** titled "Total Sales, First City, Total Orders and Total Sales by City," plotting bubble markers across Noida, Guwahati, Ranchi, Vadodara, Thane, and Mangalore, sized by sales volume.

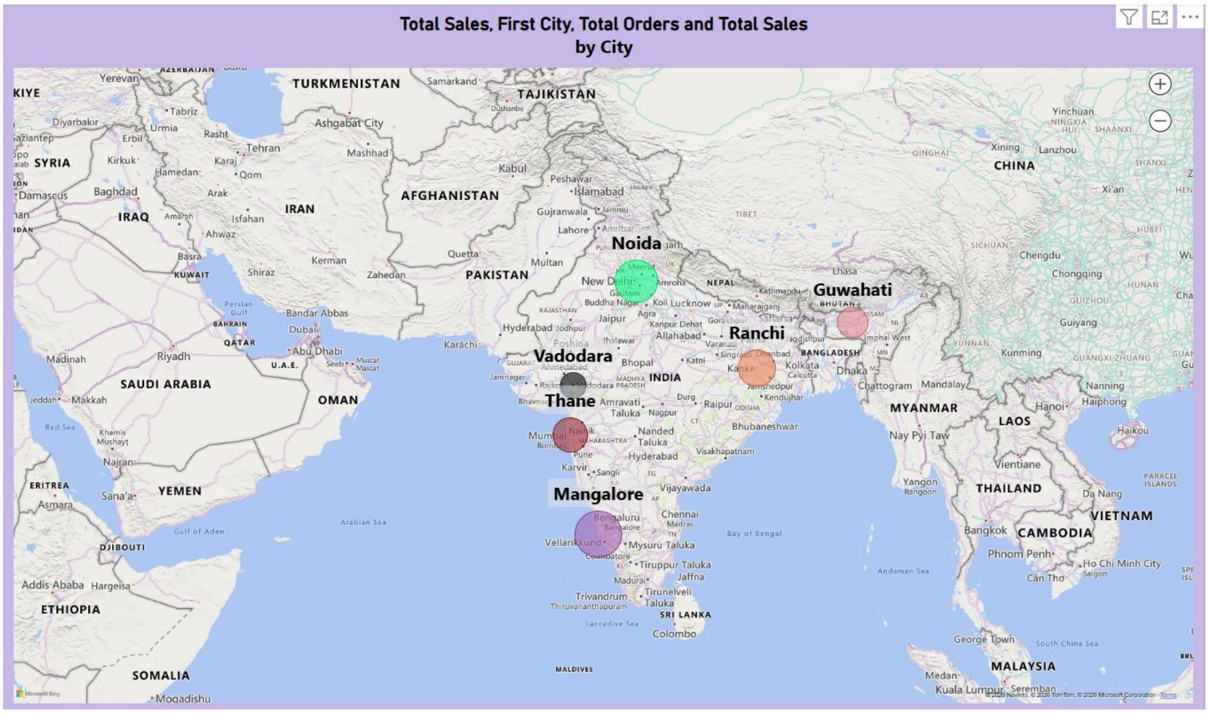

Fig — Power BI map visual showing Total Sales, Total Orders, and First City distributed across store locations in India.

Output:

The final Power BI dashboard provides:

- Total Sales, Total Orders, Total Quantity, and Total Customers at a glance via KPI cards.
- City-wise sales comparison (Mangalore is the top-performing city).
- Category-wise sales breakdown (Appliances is the best-selling category).
- Date-wise sales trend across Sep–Nov 2026.
- A geographic map view of sales and orders by city.
- Interactive filtering via Month Year, Category, and City slicers.

Conclusion

This project successfully integrated retail sales data from MySQL and customer data from a flat file into Power BI, transforming it into actionable business insights. Database design, data cleaning, relationship modeling, and dashboard development were applied to build a centralized, accurate reporting solution combining transactional and demographic data.

Learning Outcomes

Through this project, I learned:

- Designing and populating a relational database in MySQL.
- Connecting Power BI to both MySQL and flat-file sources.
- Data cleaning and transformation using Power Query.
- Building relationships across multiple tables, including a fact-to-fact link (Sales–Customer via SaleID).
- Writing DAX measures and creating KPI cards, charts, and slicers.
- Deriving business and customer insights from combined transactional data.